

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester I Examination, 2023-24

Paper No. CHB-101: Structure & Bonding and Organic Chemistry-I

Time: 3 Hours

Maximum Marks: 70

Instructions: Answer five questions in total, including Question 1 from each section which is compulsory. Use separate answer books for Section A and Section B.

SECTION A

(Structure and Bonding – Marks: 35)

Question 1. Answer all parts of the following:

[5 × 2 = 10]

- (a) What do the '+' and '-' signs in the boundary surface diagrams of orbitals indicate?
- (b) What is the physical significance of the radial wave function and the radial distribution function?
- (c) Mention the limitations of the radius ratio rules in brief.
- (d) NH₃ and NF₃ are pyramidal in shape, but the bond angle around the nitrogen center is different in the two cases. Why?
- (e) What inferences can be obtained from the knowledge of HOMO and LUMO in molecules?

Question 2.

- (a) Draw the radial distribution function versus r for 3s, 3p, and 3d orbitals, clearly depicting the salient differences in all three cases. [6]
- (b) Discuss the band theory of solids in brief. Explain conductors, insulators, and semiconductors using band gaps. [6]

Question 3.

- (a) Write the expression for the Laplacian operator in Cartesian coordinates. How are Cartesian coordinates related to polar coordinates in the Schrödinger wave equation? [6]
- (b) Explain the defects/limitations of simple Molecular Orbital Theory by drawing the energy level diagram of CO using homonuclear diatomic concepts. Draw the correct MO diagram for CO. [6]

Question 4.

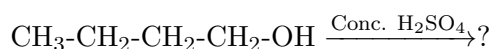
- (a) Write the molecular orbital configuration and draw the molecular orbital energy level diagram of NO. Comment on the change in bond order and magnetic behavior when NO is oxidized to NO⁺. [6]
- (b) Derive the limiting radius ratio (r^+/r^-) for tetrahedral ionic structures (coordination number 4). (Given: $\cos 30^\circ \approx 0.866$, $\sin 54^\circ 44' \approx 0.8164$). [6]

SECTION B

(Organic Chemistry-I – Marks: 35)

Question 5. Answer all parts of the following: [5 × 2 = 10]

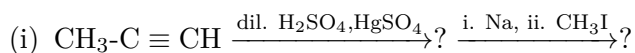
- (a) Explain why neopentyl bromide undergoes an elimination reaction even though it does not have a β -hydrogen.
- (b) Write down the Newman projections of *n*-butane having the most and least torsional strain.
- (c) Give the name and structure of the initiators used in anionic and cationic polymerization.
- (d) Complete the following reaction showing the mechanism:



- (e) Why does a polar protic solvent decrease the rate of an $\text{S}_{\text{N}}2$ reaction?
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Question 6.

- (a) Complete the following reactions, giving a suitable mechanism: [2 × 4 = 8]



- (b) Predict the *R/S* or *E/Z* nomenclature of the following molecules where applicable: [4]

- (i) Chiral center in 2-bromobutane

- (ii) $\text{Ph}(\text{CH}_3)\text{C}=\text{C}(\text{Br})\text{H}$

- (iii) $\text{CH}_3(\text{Cl})\text{C}=\text{C}(\text{Br})\text{I}$

- (iv) Chiral center in 1-chloro-2-propanol

Question 7.

- (a) How do you prepare the following compounds from phenylmagnesium iodide (PhMgI)? Write down stepwise mechanisms: [6]

- (i) Benzoic acid

- (ii) Bromobenzene

- (iii) Phenol

- (b) How do you distinguish primary and secondary alcohols by the Lucas test (substitution method)? [3]
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- (c) How do you prepare a 1,3-diketone from ethyl acetoacetate? [3]
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Question 8.

- (a) How do you prepare the following compounds from diethyl malonate? [6]
- (i) Glutaric acid
 - (ii) Barbituric acid
- (b) Predict the major products of the following reactions: [4]
- (i) $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-Br} + \text{EtSNa} \xrightarrow{\text{EtOH}} ?$
 - (ii) $\text{CH}_2=\text{CH-CH=CH}_2 + \text{Ethylene} \xrightarrow{100^\circ\text{C}} ?$
- (c) How do you prepare adipic acid from cyclohexene? [2]